



UNITED INTERNATIONAL UNIVERSITY (UIU)

Dept. of Electrical & Electronic Engineering (EEE)

Final Project, Trimester: Summer, Year: 2026

Course: EEE 3310 (Sec: A, B)

Title: Digital Signal Processing Laboratory

Full Marks: **40**

Problem Statement:

1. Analyze a real-life complex engineering problem and design and implement the solution of it using signal processing tool (e.g., MATLAB) [CO1-PO2, CO2-PO3].
 - a. The solution must include considerable amount of digital signal processing knowledge including at least a few of the following:
 - i. Audio Signal as Input
 - ii. Multipath channel effects and channel parameters
 - iii. Equalizer
 - b. You can use any kind of **equalizer designing process and coding tool** providing that the project fulfills the requirement 1a. One example of an equalizer that can be used here is the zero-forcing equalizer.
2. You must work in a group to solve the problem. Group work and individual contribution – both will be judged. [CO4-PO9]
3. Prepare and submit a technical report on the project [CO3-PO10].
4. Demonstrate the project to the peers and the instructor [CO3-PO10].
5. Present the project by computer presentation to the peers and the instructor [CO3-PO10].

Timeline:

- September 23, 27 (B, A), 2026: Demonstration of the channel estimation part of the project.
- September 29 (B) and October 3 (A), 2026: Demonstration of the final results and analysis, presentation using slides and report submission.
- October 5, 2026 (Monday): Participate in the Project Show.

Report:

- The report should include the following:
 1. Title page
 2. Introduction
 3. Description of the project
 4. Full flow chart and explanation
 5. Results
 6. Conclusion

Definition of Complex Engineering Problem:

Complex Engineering Problems have characteristic P1 and some or all of P2 to P7**:

P	Attribute	Statement
P1	Depth of knowledge required	Cannot be resolved without in-depth engineering knowledge at the level of one or more of K3-K6 or K8 which allows a fundamentals-based, first principles analytical approach
P2	Range of conflicting requirements	Involve wide-ranging or conflicting technical, engineering and other issues
P3	Depth of analysis required	Have no obvious solution and require abstract thinking, originality in analysis to formulate suitable models
P4	Familiarity of issues	Involve infrequently encountered issues
P5	Extent of applicable codes	Are outside problems encompassed by standards and codes of practice for professional engineering
P6	Extent of stakeholder involvement and conflicting requirements	Involve diverse groups of stakeholders with widely varying needs
P7	Interdependence	Are high-level problems including many component parts or sub-problems

This project aims to include **P1, P2, P3

CO and PO mapping:

CO	Description of the CO	Corr. PO	Description of the PO
CO1	Analyze behavior / response of electronic circuits using modern tools (PO2).	PO2	Problem analysis: Identify, formulate, research the literature and analyze complex engineering problems and reach substantiated conclusions using first principles of mathematics, the natural sciences and the engineering sciences.
CO2	Design and implement electronic circuits meeting specific requirements (PO3).	PO3	Design/development of solutions: Design solutions for complex engineering problems and design system components or processes that meet the specified needs.
CO3	Prepare and present a technical report (PO10).	PO10	Communication: Communicate effectively about complex engineering activities with the engineering community and with society at large. Be able to

			comprehend and write effective reports, design documentation, make effective presentations and give and receive clear instructions.
CO4	Work individually and in a group to complete a project (PO9).	PO9	Individual work and teamwork: Function effectively as an individual, and as a member or leader in diverse teams and in multi-disciplinary settings.

Grading Rubric:

	Course Outcome	Criterion	K	P	A
Initial presentation (5)	CO1-PO2		K3, K8	P1, P2	
Equalizer Design and Coefficient Calculation (10)	CO2-PO3 CO4-PO9	Project should mention the channel parameters and equalizer parameters for Zero-Forcing equalizer.	K5	P2, P3	
ISI removal or reduction from received signal (10)	CO2-PO3 CO4-PO9	Project should be able to remove echo/ISI effect from the received audio signal.	K5	P2, P3	
Individual viva (5)	CO4-PO9	Questions regarding the theory and implementation will be asked to find out the individual contribution.	K3, K5		
Presentation (5)	CO3-PO10	Criteria for the report will be followed. Individual presentation skills will be judged as well.			A1, A2
Report (5)	CO3-PO10	Title page and overall representation			
		Introduction and Conclusion (2)			
		Description of the project (2)			A1
		Full flow-chart/algorithm of code and explanation (2)			A2
		Plot of channel and equalizer (2)			
		Result (2)			

Project: Reduce the Inter-Symbol Interference (ISI) by designing a Zero-Forcing Equalizer

A common channel distortion that arises in a communication channel is that of a multipath channel distortion due to the reflections at certain points in the channel. This results in Inter-Symbol Interference (ISI) that can degrade the communication severely. A typical everyday example of this can be found in echoes that experience either outside or in an indoor large hall room as shown in Figure 1.

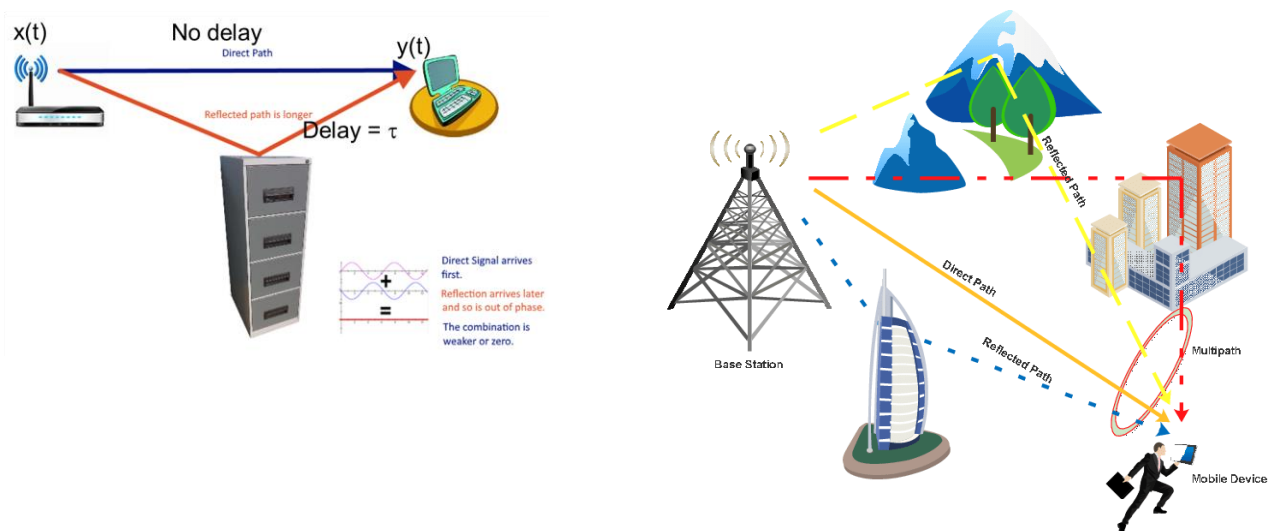


Figure 1. Illustration of multipath reflection in signal propagation

This project aims to demonstrate the ISI effect in an audio signal and design an equalizer to reduce the degradation to a satisfactory level. An equalizer can be thought of an additional system that is used after the channel similar to pole-zero cancellation where an additional system is introduced after a main system (see Figure 2).

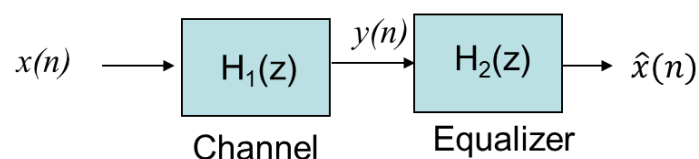


Figure 2. Use of an Equalizer to reduce channel distortion

The solution comprises a two-stage approach: (i) Estimation of the channel parameters through channel sounding (ii) Estimating the coefficients of the equalizer filter. Two data files will be provided in this respect: (i) output from channel sounding stage and (ii) distorted audio file due to ISI effects from the channel.

Objectives

1. To model a communication channel that introduces ISI into a transmitted signal
2. To derive the Zero-Forcing Equalizer coefficients by computing the inverse of the channel's transfer function in the z-domain.

3. To implement the Zero-Forcing Equalizer as an IIR/FIR Filter in MATLAB and apply it to remove ISI from the received signal
4. To design the equalizer with an optimal filter length to achieve effective ISI cancellation while maintaining computational efficiency

Specifications (P1–P3)

1. **P1 – Knowledge:** Use digital signal processing concepts to design the zero-forcing equalizer.
2. **P1 – Analysis:** Calculate the input and output coefficient to analyze the channel condition.
3. **P2 – Conflict:** The channel and equalizer must be designed in such a way that the system remains stable and the system poles converges within the unit circle.
4. **P3 – Original Design:** Design and test the channel condition with an audio signal to analyze the echo cancellation of a channel.

Hints

To design an equalizer $E(z)$ you can use the following equation $E(z) = \frac{1}{H(z)}$.

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